**FIG. 2.63**

Determining the PIV level for the diodes of the CT transformer full-wave rectifier.

PIV The network of Fig. 2.63 will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as established by the adjoining loop results in

$$\begin{aligned} \text{PIV} &= V_{\text{secondary}} + V_R \\ &= V_m + V_m \end{aligned}$$

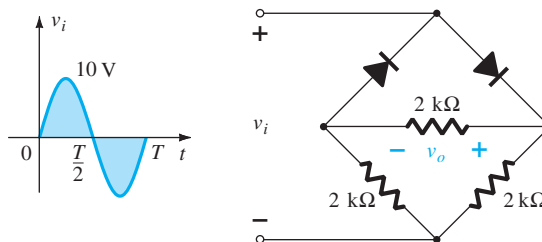
and

$$\text{PIV} \cong 2V_m$$

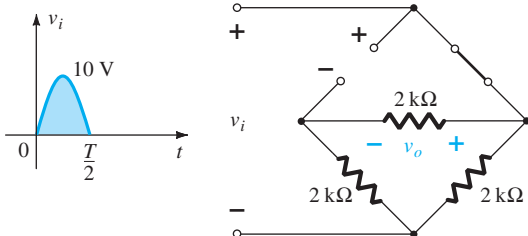
CT transformer, full-wave rectifier

(2.13)

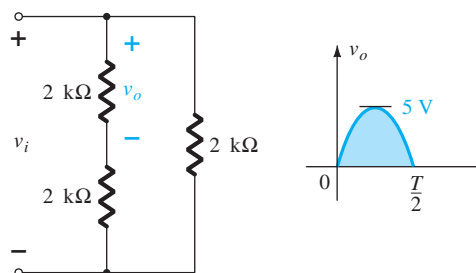
EXAMPLE 2.17 Determine the output waveform for the network of Fig. 2.64 and calculate the output dc level and the required PIV of each diode.

**FIG. 2.64**

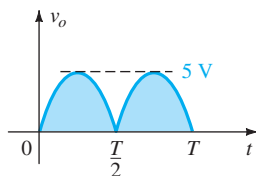
Bridge network for Example 2.17.

**FIG. 2.65**

Network of Fig. 2.64 for the positive region of v_i .

**FIG. 2.66**

Redrawn network of Fig. 2.65.

**FIG. 2.67**

Resulting output for Example 2.17.

Solution: The network appears as shown in Fig. 2.65 for the positive region of the input voltage. Redrawing the network results in the configuration of Fig. 2.66, where $v_o = \frac{1}{2}v_i$ or $V_{o\text{max}} = \frac{1}{2}V_{i\text{max}} = \frac{1}{2}(10 \text{ V}) = 5 \text{ V}$, as shown in Fig. 2.66. For the negative part of the input, the roles of the diodes are interchanged and v_o appears as shown in Fig. 2.67.

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

$$V_{\text{dc}} = 0.636(5 \text{ V}) = 3.18 \text{ V}$$

or that available from a half-wave rectifier with the same input. However, the PIV as determined from Fig. 2.59 is equal to the maximum voltage across R , which is 5 V, or half of that required for a half-wave rectifier with the same input.

2.8 CLIPPERS

The previous section on rectification gives clear evidence that diodes can be used to change the appearance of an applied waveform. This section on clippers and the next on clampers will expand on the wave-shaping abilities of diodes.

Clippers are networks that employ diodes to “clip” away a portion of an input signal without distorting the remaining part of the applied waveform.

The half-wave rectifier of Section 2.6 is an example of the simplest form of diode clipper—one resistor and a diode. Depending on the orientation of the diode, the positive or negative region of the applied signal is “clipped” off.

There are two general categories of clippers: *series* and *parallel*. The series configuration is defined as one where the diode is in series with the load, whereas the parallel variety has the diode in a branch parallel to the load.

Series

The response of the series configuration of Fig. 2.68a to a variety of alternating waveforms is provided in Fig. 2.68b. Although first introduced as a half-wave rectifier (for sinusoidal waveforms), there are no boundaries on the type of signals that can be applied to a clipper.

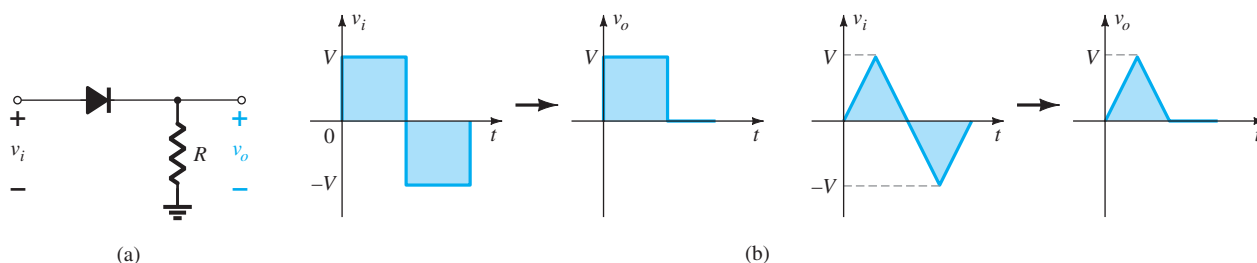


FIG. 2.68
Series clipper.

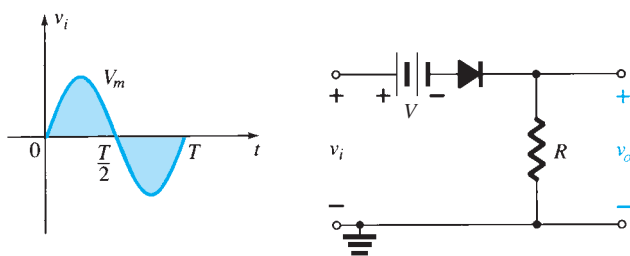


FIG. 2.69
Series clipper with a dc supply.

The addition of a dc supply to the network as shown in Fig. 2.69 can have a pronounced effect on the analysis of the series clipper configuration. The response is not as obvious because the dc supply can aid or work against the source voltage, and the dc supply can be in the leg between the supply and output or in the branch parallel to the output.

There is no general procedure for analyzing networks such as the type in Fig. 2.69, but there are some things one can do to give the analysis some direction.

First and most important:

- 1. Take careful note of where the output voltage is defined.**

In Fig. 2.69 it is directly across the resistor R . In some cases it may be across a combination of series elements.

Next:

- 2. Try to develop an overall sense of the response by simply noting the “pressure” established by each supply and the effect it will have on the conventional current direction through the diode.**

In Fig. 2.69, for instance, any positive voltage of the supply will try to turn the diode on by establishing a conventional current through the diode that matches the arrow in the diode symbol. However, the added dc supply V will oppose that applied voltage and try to keep the diode in the “off” state. The result is that any supply voltage greater than V volts will turn the diode on and conduction can be established through the load resistor. Keep in mind that we are dealing with an ideal diode for the moment, so the turn-on voltage is simply 0 V. In general, therefore, for the network of Fig. 2.69 we can conclude that the

diode will be on for any voltage v_i that is greater than V volts and off for any lesser voltage. For the “off” condition, the output would be 0 V due to the lack of current, and for the “on” condition it would simply be $v_o = v_i - V$ as determined by Kirchhoff’s voltage law.

3. Determine the applied voltage (transition voltage) that will result in a change of state for the diode from the “off” to the “on” state.

This step will help to define a region of the applied voltage when the diode is on and when it is off. On the characteristics of an ideal diode this will occur when $V_D = 0$ V and $I_D = 0$ mA. For the approximate equivalent this is determined by finding the applied voltage when the diode has a drop of 0.7 V across it (for silicon) and $I_D = 0$ mA.

This exercise was applied to the network of Fig. 2.69 as shown in Fig. 2.70. Note the substitution of the short-circuit equivalent for the diode and the fact that the voltage across the resistor is 0 V because the diode current is 0 mA. The result is $v_i - V = 0$, and so

$$v_i = V \quad (2.14)$$

is the transition voltage.

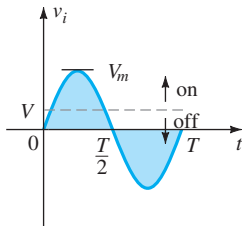


FIG. 2.71

Using the transition voltage to define the “on” and “off” regions.

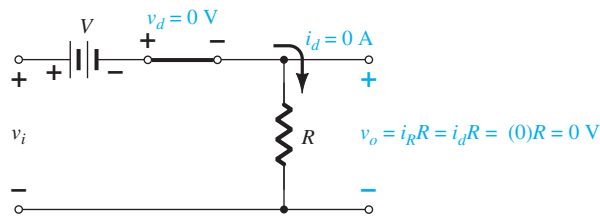


FIG. 2.70

Determining the transition level for the circuit of Fig. 2.69.

This permits drawing a line on the sinusoidal supply voltage as shown in Fig. 2.71 to define the regions where the diode is on and off.

For the “on” region, as shown in Fig. 2.72, the diode is replaced by a short-circuit equivalent, and the output voltage is defined by

$$v_o = v_i - V \quad (2.15)$$

For the “off” region, the diode is an open circuit, $I_D = 0$ mA, and the output voltage is

$$v_o = 0 \text{ V}$$

4. It is often helpful to draw the output waveform directly below the applied voltage using the same scales for the horizontal axis and the vertical axis.

Using this last piece of information, we can establish the 0-V level on the plot of Fig. 2.73 for the region indicated. For the “on” condition, Eq. (2.15) can be used to find the output voltage when the applied voltage has its peak value:

$$v_{o\text{peak}} = V_m - V$$

and this can be added to the plot of Fig. 2.73. It is then simple to fill in the missing section of the output curve.

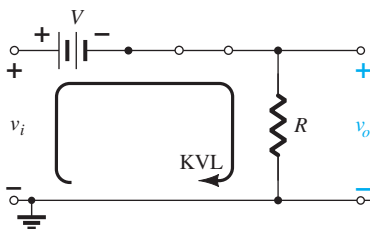


FIG. 2.72

Determining v_o for the diode in the “on” state.

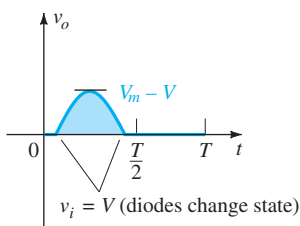


FIG. 2.73

Sketching the waveform of v_o using the results obtained for v_o above and below the transition level.

EXAMPLE 2.18 Determine the output waveform for the sinusoidal input of Fig. 2.74.

Solution:

Step 1: The output is again directly across the resistor R .

Step 2: The positive region of v_i and the dc supply are both applying “pressure” to turn the diode on. The result is that we can safely assume the diode is in the “on” state for the entire range of positive voltages for v_i . Once the supply goes negative, it would have to exceed the dc supply voltage of 5 V before it could turn the diode off.

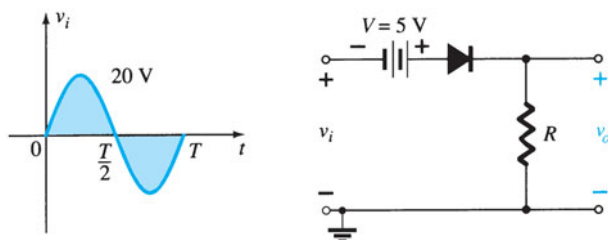


FIG. 2.74

Series clipper for Example 2.18.

Step 3: The transition model is substituted in Fig. 2.75, and we find that the transition from one state to the other will occur when

$$v_i + 5 \text{ V} = 0 \text{ V}$$

or

$$v_i = -5 \text{ V}$$

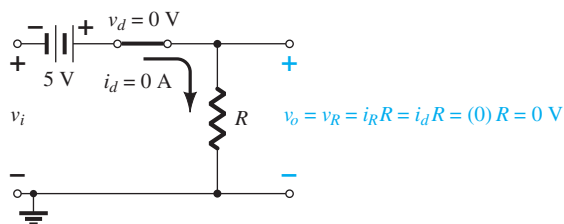


FIG. 2.75

Determining the transition level for the clipper of Fig. 2.74.

Step 4: In Fig. 2.76 a horizontal line is drawn through the applied voltage at the transition level. For voltages less than -5 V the diode is in the open-circuit state and the output is 0 V , as shown in the sketch of v_o . Using Fig. 2.76, we find that for conditions when the diode is on and the diode current is established the output voltage will be the following, as determined using Kirchhoff's voltage law:

$$v_o = v_i + 5 \text{ V}$$

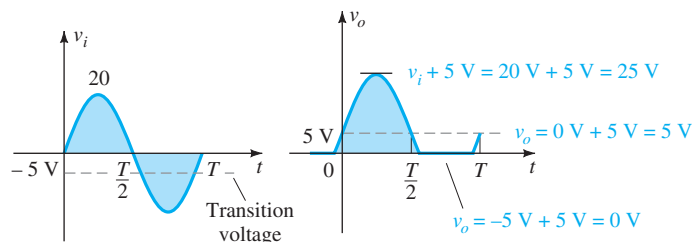


FIG. 2.76

Sketching v_o for Example 2.18.

The analysis of clipper networks with square-wave inputs is actually easier than with sinusoidal inputs because only two levels have to be considered. In other words, the network can be analyzed as if it had two dc level inputs with the resulting v_o plotted in the proper time frame. The next example demonstrates the procedure.

EXAMPLE 2.19 Find the output voltage for the network examined in Example 2.18 if the applied signal is the square wave of Fig. 2.77.

Solution: For $v_i = 20 \text{ V}$ ($0 \rightarrow T/2$) the network of Fig. 2.78 results. The diode is in the short-circuit state, and $v_o = 20 \text{ V} + 5 \text{ V} = 25 \text{ V}$. For $v_i = -10 \text{ V}$ the network of Fig. 2.79

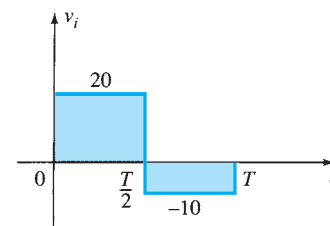


FIG. 2.77

Applied signal for Example 2.19.

results, placing the diode in the “off” state, and $v_o = i_R R = (0)R = 0$ V. The resulting output voltage appears in Fig. 2.80.

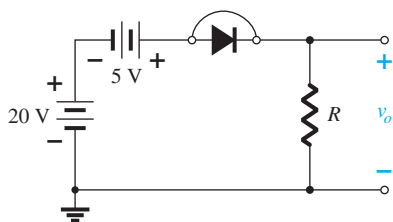


FIG. 2.78
 v_o at $v_i = +20$ V.

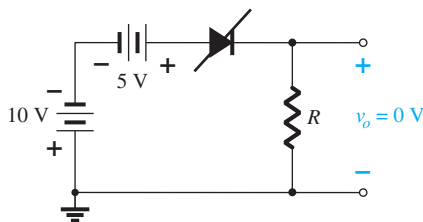


FIG. 2.79
 v_o at $v_i = -10$ V.

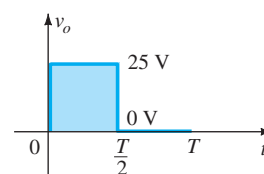


FIG. 2.80
Sketching v_o for Example 2.19.

Note in Example 2.19 that the clipper not only clipped off 5 V from the total swing, but also raised the dc level of the signal by 5 V.

Parallel

The network of Fig. 2.81 is the simplest of parallel diode configurations with the output for the same inputs of Fig. 2.68. The analysis of parallel configurations is very similar to that applied to series configurations, as demonstrated in the next example.

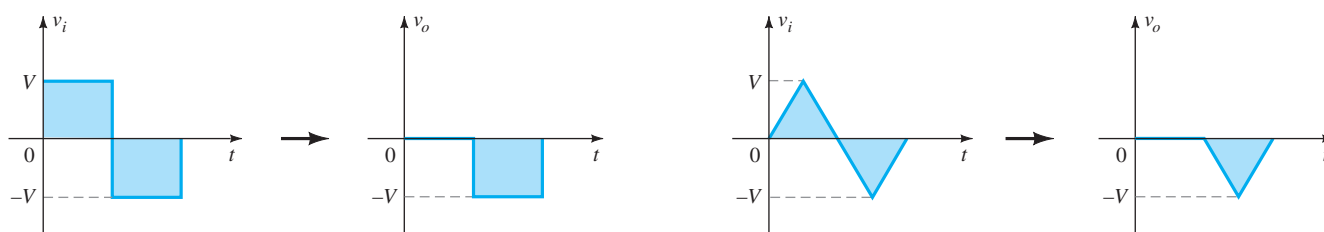
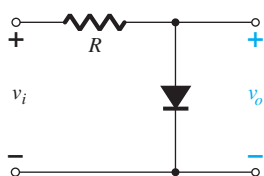


FIG. 2.81
Response to a parallel clipper.

EXAMPLE 2.20 Determine v_o for the network of Fig. 2.82.

Solution:

Step 1: In this example the output is defined across the series combination of the 4-V supply and the diode, not across the resistor R .

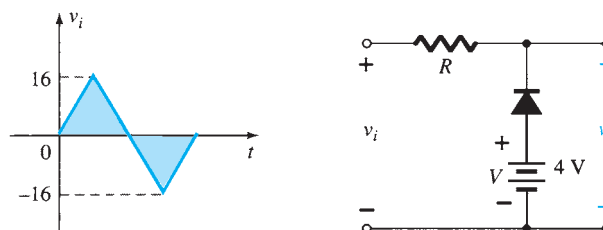


FIG. 2.82
Example 2.20.

Step 2: The polarity of the dc supply and the direction of the diode strongly suggest that the diode will be in the “on” state for a good portion of the negative region of the input signal. In fact, it is interesting to note that since the output is directly across the series combination, when the diode is in its short-circuit state the output voltage will be directly across the 4-V dc supply, requiring that the output be fixed at 4 V. In other words, when the diode is on the output will be 4 V. Other than that, when the diode is an open circuit, the current through the series network will be 0 mA and the voltage drop across the resistor will be 0 V. That will result in $v_o = v_i$ whenever the diode is off.

Step 3: The transition level of the input voltage can be found from Fig. 2.83 by substituting the short-circuit equivalent and remembering the diode current is 0 mA at the instant of transition. The result is a change in state when

$$v_i = 4 \text{ V}$$

Step 4: In Fig. 2.84 the transition level is drawn along with $v_o = 4 \text{ V}$ when the diode is on. For $v_i \geq 4 \text{ V}$, $v_o = 4 \text{ V}$, and the waveform is simply repeated on the output plot.

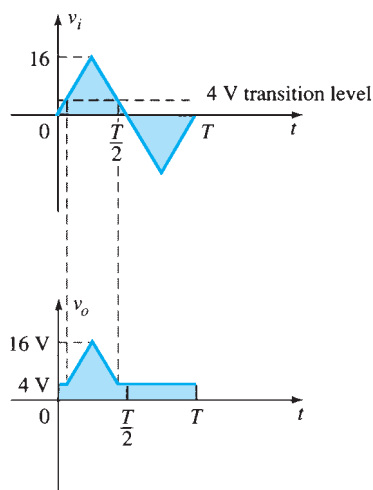


FIG. 2.84

Sketching v_o for Example 2.20.

To examine the effects of the knee voltage V_K of a silicon diode on the output response, the next example will specify a silicon diode rather than the ideal diode equivalent.

EXAMPLE 2.21 Repeat Example 2.20 using a silicon diode with $V_K = 0.7 \text{ V}$.

Solution: The transition voltage can first be determined by applying the condition $i_d = 0 \text{ A}$ at $v_d = V_D = 0.7 \text{ V}$ and obtaining the network of Fig. 2.85. Applying Kirchhoff's voltage law around the output loop in the clockwise direction, we find that

$$v_i + V_K - V = 0$$

and

$$v_i = V - V_K = 4 \text{ V} - 0.7 \text{ V} = 3.3 \text{ V}$$

For input voltages greater than 3.3 V, the diode will be an open circuit and $v_o = v_i$. For input voltages less than 3.3 V, the diode will be in the “on” state and the network of Fig. 2.86 results, where

$$v_o = 4 \text{ V} - 0.7 \text{ V} = 3.3 \text{ V}$$

The resulting output waveform appears in Fig. 2.87. Note that the only effect of V_K was to drop the transition level to 3.3 from 4 V.

There is no question that including the effects of V_K will complicate the analysis somewhat, but once the analysis is understood with the ideal diode, the procedure, including the effects of V_K , will not be that difficult.

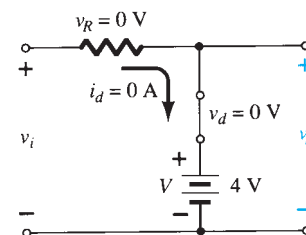


FIG. 2.83

Determining the transition level for Example 2.20.

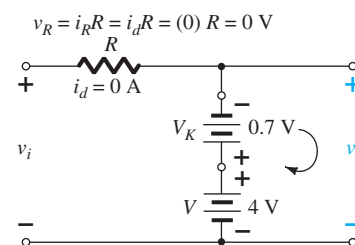


FIG. 2.85

Determining the transition level for the network of Fig. 2.82.

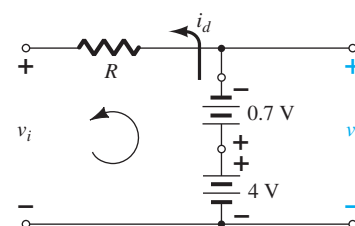


FIG. 2.86

Determining v_o for the diode of Fig. 2.82 in the “on” state.

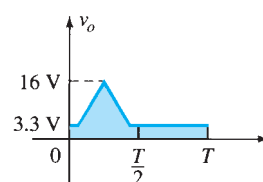
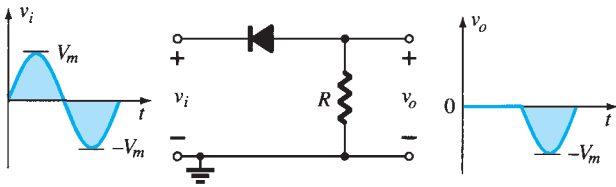


FIG. 2.87

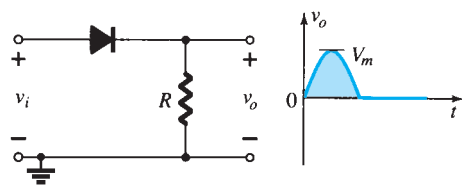
Sketching v_o for Example 2.21.

Simple Series Clippers (Ideal Diodes)

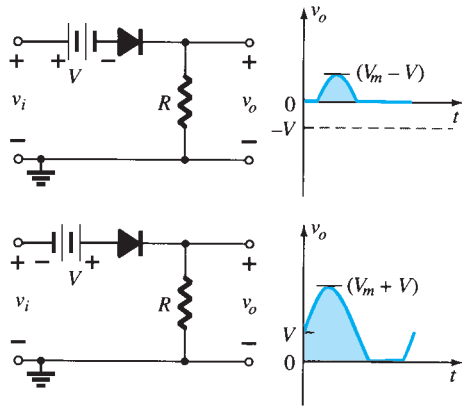
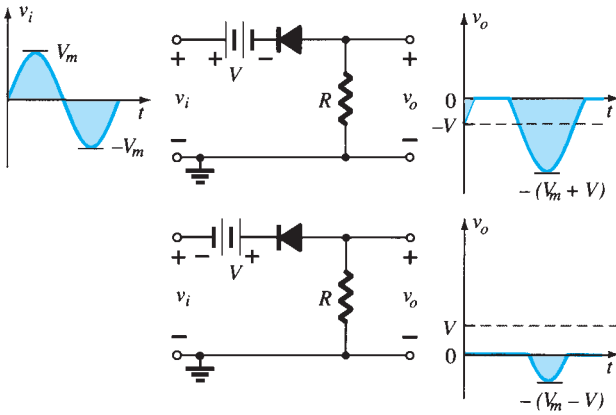
POSITIVE



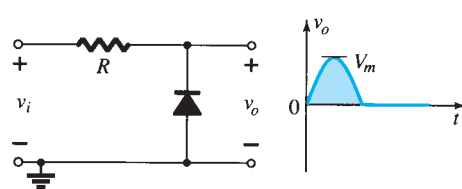
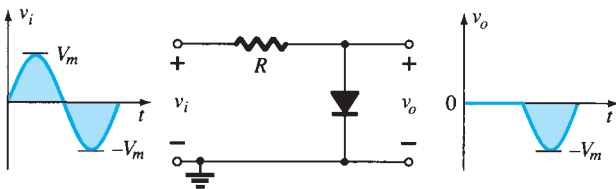
NEGATIVE



Biased Series Clippers (Ideal Diodes)



Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

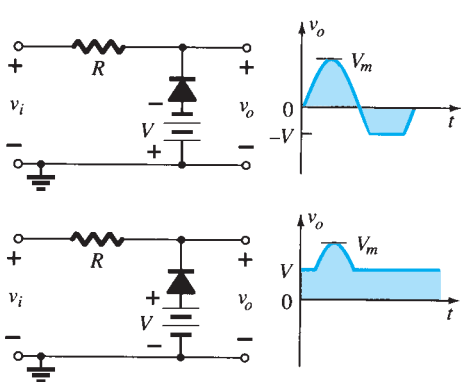
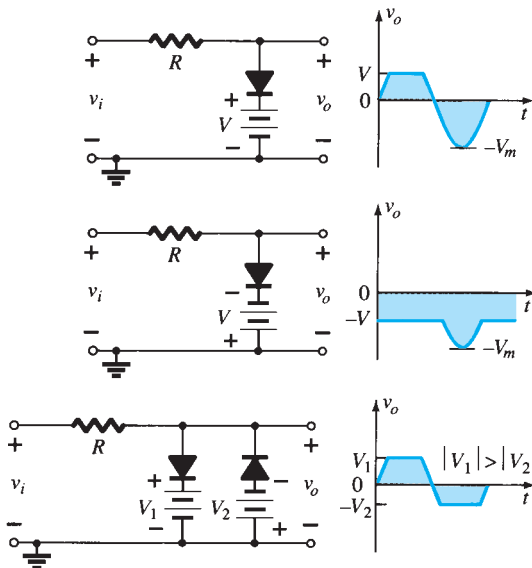


FIG. 2.88
Clipping circuits.

A variety of series and parallel clippers with the resulting output for the sinusoidal input are provided in Fig. 2.88. In particular, note the response of the last configuration, with its ability to clip off a positive and a negative section as determined by the magnitude of the dc supplies.

2.9 CLAMPERS

The previous section investigated a number of diode configurations that clipped off a portion of the applied signal without changing the remaining part of the waveform. This section will examine a variety of diode configurations that shift the applied signal to a different level.

A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal.

Additional shifts can also be obtained by introducing a dc supply to the basic structure. The chosen resistor and capacitor of the network must be chosen such that the time constant determined by $\tau = RC$ is sufficiently large to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is nonconducting. Throughout the analysis we assume that for all practical purposes the capacitor fully charges or discharges in five time constants.

The simplest of clamper networks is provided in Fig. 2.89. It is important to note that the capacitor is connected directly between input and output signals and the resistor and the diode are connected in parallel with the output signal.

Clamping networks have a capacitor connected directly from input to output with a resistive element in parallel with the output signal. The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

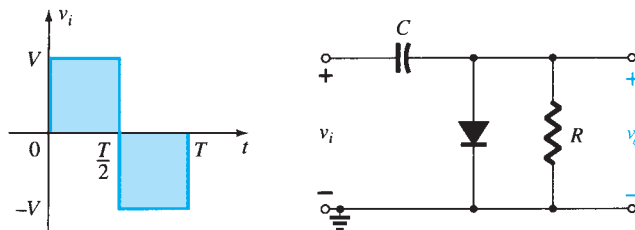


FIG. 2.89
Clamper.

There is a sequence of steps that can be applied to help make the analysis straightforward. It is not the only approach to examining clammers, but it does offer an option if difficulties surface.

Step 1: Start the analysis by examining the response of the portion of the input signal that will forward bias the diode.

Step 2: During the period that the diode is in the “on” state, assume that the capacitor will charge up instantaneously to a voltage level determined by the surrounding network.

For the network of Fig. 2.89 the diode will be forward biased for the positive portion of the applied signal. For the interval 0 to $T/2$ the network will appear as shown in Fig. 2.90. The short-circuit equivalent for the diode will result in $v_o = 0$ V for this time interval, as shown in the sketch of v_o in Fig. 2.92. During this same interval of time, the time constant determined by $\tau = RC$ is very small because the resistor R has been effectively “shorted out” by the conducting diode and the only resistance present is the inherent (contact, wire) resistance of the network. The result is that the capacitor will quickly charge to the peak value of V volts as shown in Fig. 2.90 with the polarity indicated.

Step 3: Assume that during the period when the diode is in the “off” state the capacitor holds on to its established voltage level.

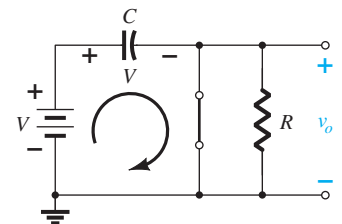


FIG. 2.90
Diode “on” and the capacitor charging to V volts.

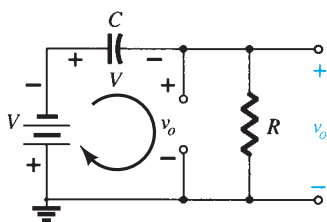


FIG. 2.91

Determining v_o with the diode “off.”

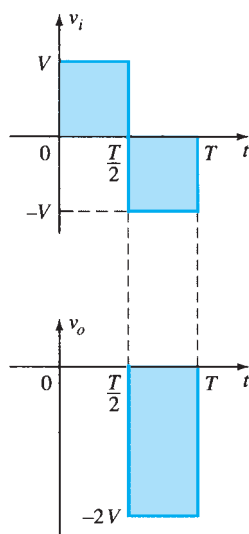


FIG. 2.92

Sketching v_o for the network of Fig. 2.91.

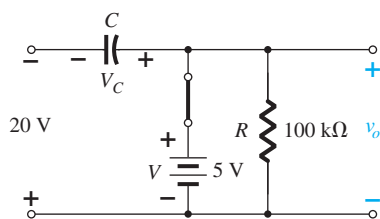


FIG. 2.94

Determining v_o and V_C with the diode in the “on” state.

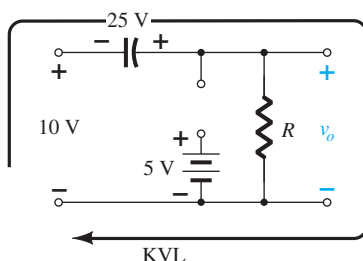


FIG. 2.95

Determining v_o with the diode in the “off” state.

Step 4: Throughout the analysis, maintain a continual awareness of the location and defined polarity for v_o to ensure that the proper levels are obtained.

When the input switches to the $-V$ state, the network will appear as shown in Fig. 2.91, with the open-circuit equivalent for the diode determined by the applied signal and stored voltage across the capacitor—both “pressuring” current through the diode from cathode to anode. Now that R is back in the network the time constant determined by the RC product is sufficiently large to establish a discharge period 5τ , much greater than the period $T/2 \rightarrow T$, and it can be assumed on an approximate basis that the capacitor holds onto all its charge and, therefore, voltage (since $V = Q/C$) during this period.

Since v_o is in parallel with the diode and resistor, it can also be drawn in the alternative position shown in Fig. 2.91. Applying Kirchhoff’s voltage law around the input loop results in

$$-V - V - v_o = 0$$

and

$$v_o = -2V$$

The negative sign results from the fact that the polarity of $2V$ is opposite to the polarity defined for v_o . The resulting output waveform appears in Fig. 2.92 with the input signal. The output signal is clamped to 0 V for the interval 0 to $T/2$ but maintains the same total swing ($2V$) as the input.

Step 5: Check that the total swing of the output matches that of the input.

This is a property that applies for all clamping networks, giving an excellent check on the results obtained.

EXAMPLE 2.22 Determine v_o for the network of Fig. 2.93 for the input indicated.

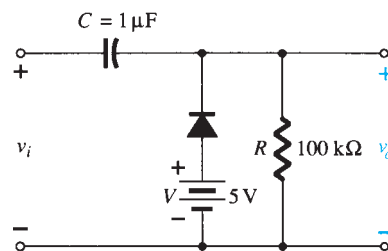
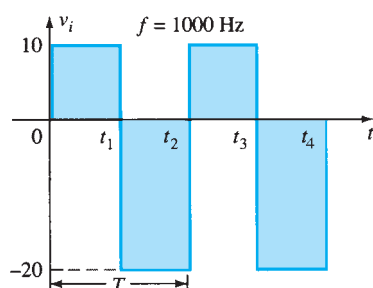


FIG. 2.93

Applied signal and network for Example 2.22.

Solution: Note that the frequency is 1000 Hz, resulting in a period of 1 ms and an interval of 0.5 ms between levels. The analysis will begin with the period $t_1 \rightarrow t_2$ of the input signal since the diode is in its short-circuit state. For this interval the network will appear as shown in Fig. 2.94. The output is across R , but it is also directly across the 5-V battery if one follows the direct connection between the defined terminals for v_o and the battery terminals. The result is $v_o = 5$ V for this interval. Applying Kirchhoff’s voltage law around the input loop results in

$$-20 \text{ V} + V_C - 5 \text{ V} = 0$$

and

$$V_C = 25 \text{ V}$$

The capacitor will therefore charge up to 25 V. In this case the resistor R is not shorted out by the diode, but a Thévenin equivalent circuit of that portion of the network that includes the battery and the resistor will result in $R_{Th} = 0 \Omega$ with $E_{Th} = V = 5$ V. For the period $t_2 \rightarrow t_3$ the network will appear as shown in Fig. 2.95.

The open-circuit equivalent for the diode removes the 5-V battery from having any effect on v_o , and applying Kirchhoff’s voltage law around the outside loop of the network results in

$$+10 \text{ V} + 25 \text{ V} - v_o = 0$$

and

$$v_o = 35 \text{ V}$$

The time constant of the discharging network of Fig. 2.95 is determined by the product RC and has the magnitude

$$\tau = RC = (100 \text{ k}\Omega)(0.1 \text{ }\mu\text{F}) = 0.01 \text{ s} = 10 \text{ ms}$$

The total discharge time is therefore $5\tau = 5(10 \text{ ms}) = 50 \text{ ms}$.

Since the interval $t_2 \rightarrow t_3$ will only last for 0.5 ms, it is certainly a good approximation that the capacitor will hold its voltage during the discharge period between pulses of the input signal. The resulting output appears in Fig. 2.96 with the input signal. Note that the output swing of 30 V matches the input swing as noted in step 5.

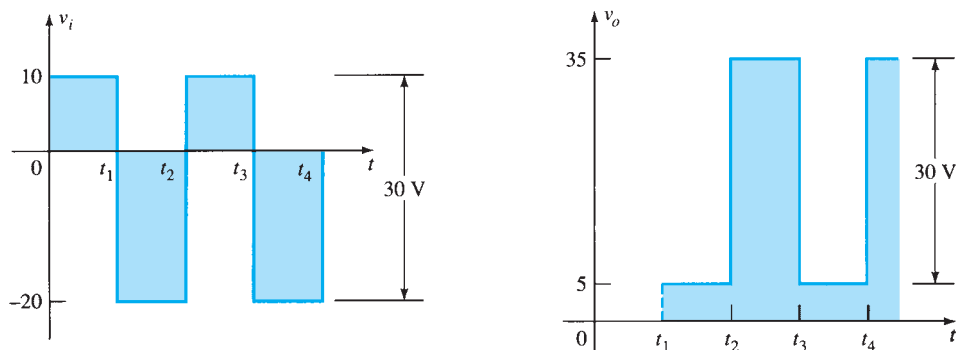


FIG. 2.96

v_i and v_o for the clamper of Fig. 2.93.

EXAMPLE 2.23 Repeat Example 2.22 using a silicon diode with $V_K = 0.7 \text{ V}$.

Solution: For the short-circuit state the network now takes on the appearance of Fig. 2.97, and v_o can be determined by Kirchhoff's voltage law in the output section:

$$+5 \text{ V} - 0.7 \text{ V} - v_o = 0$$

and

$$v_o = 5 \text{ V} - 0.7 \text{ V} = 4.3 \text{ V}$$

For the input section Kirchhoff's voltage law results in

$$-20 \text{ V} + V_C + 0.7 \text{ V} - 5 \text{ V} = 0$$

and

$$V_C = 25 \text{ V} - 0.7 \text{ V} = 24.3 \text{ V}$$

For the period $t_2 \rightarrow t_3$ the network will now appear as in Fig. 2.98, with the only change being the voltage across the capacitor. Applying Kirchhoff's voltage law yields

$$+10 \text{ V} + 24.3 \text{ V} - v_o = 0$$

and

$$v_o = 34.3 \text{ V}$$

The resulting output appears in Fig. 2.99, verifying the statement that the input and output swings are the same.

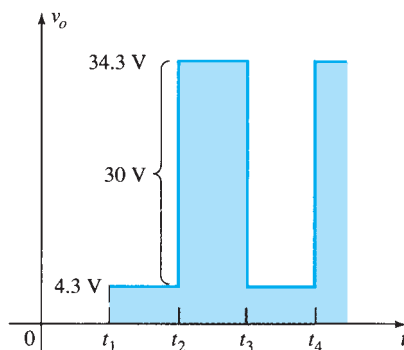


FIG. 2.99

Sketching v_o for the clamper of Fig. 2.93 with a silicon diode.

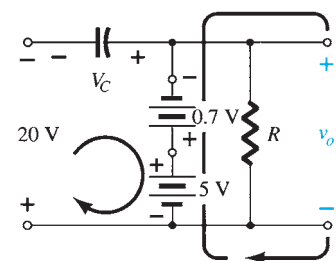


FIG. 2.97

Determining v_o and V_C with the diode in the "on" state.

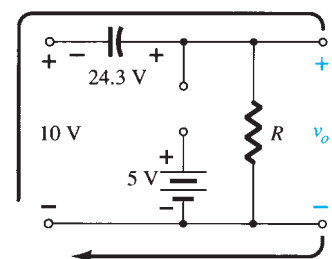


FIG. 2.98

Determining v_o with the diode in the open state.